

CME 307 / MS&E 311 / OIT 676: Optimization

Interior Point Methods

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Management Science and Engineering
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slides developed with Prof. Luiz-Rafael Santos, UFSC <https://lrsantos11.github.io/>

Outline

IPM for linear and quadratic programs

IPM for convex nonlinear programming

IPM for conic optimization

IPM for linear and quadratic programs

Linear/Quadratic Program

$$\begin{array}{ll}\text{minimize} & c^\top x + \frac{1}{2}x^\top Qx \\ \text{subject to} & Ax = b, \\ & x \geq 0,\end{array}$$

where $Q \in \mathbf{S}_+^n$, and $A \in \mathbb{R}^{m \times n}$ is full-rank.

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Simplex: vertex to vertex
IPM: go through the middle!



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Advantages of vertex solution vs interior solution?

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IPM: go through the middle!



Building blocks of IPM

Ingredients for Interior Point Method

- ▶ Duality theory: Lagrangian function; KKT (first order optimality) condition.
- ▶ Barrier function: logarithmic barrier.
- ▶ Newton's method (and a good linear solver)

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The reward: fantastic convergence properties!

- ▶ Theoretical: $O(\sqrt{n} \log(1/\varepsilon))$ iterations
- ▶ Practical: $O(\log n \log(1/\varepsilon))$ iterations

(but the per-iteration cost may be high due to the Newton solve: often $O(n^3)$)

IPM: algorithmic template

IPM procedure

- ▶ replace inequalities with log barriers;
- ▶ form the Lagrangian;
- ▶ write down the KKT conditions of the perturbed problem;
- ▶ find one (or more) directions using [Newton's method](#) on the KKT system;
- ▶ (decide how to combine the directions and) compute a stepsize.

Duality and KKT conditions

Primal-dual QPs

Primal problem

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Dual problem

$$\begin{array}{ll}\text{maximize} & b^\top y - \frac{1}{2}x^\top Qx \\ \text{subject to} & A^\top y + s - Qx = c \\ & s \geq 0\end{array}$$

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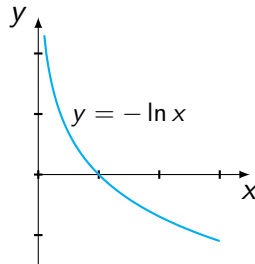
KKT conditions

$$\begin{array}{ll}Ax = b & \triangleright (\text{primal feasibility}) \\ A^\top y + s - Qx = c & \triangleright (\text{dual feasibility}) \\ XS\mathbf{1} = 0 & \triangleright (\text{complementarity: } x_i s_i = 0, i = 1, \dots, n) \\ (x, s) \geq 0\end{array}$$

where $X = \mathbf{diag}(x_1, \dots, x_n)$, $S = \mathbf{diag}(s_1, \dots, s_n) \in \mathbb{R}^{n \times n}$, and $e = (1, \dots, 1) \in \mathbb{R}^n$.

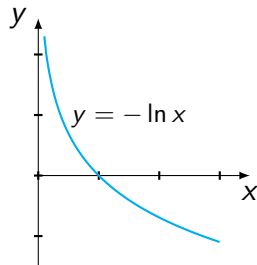
Logarithmic barrier

$-\ln x_j$
replaces the inequality
 $x_j \geq 0$



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$$\text{minimize} \quad -\sum_{j=1}^n \ln x_j \quad \Longleftrightarrow \quad \text{maximize} \quad \prod_{1 \leq j \leq n} x_j$$

$\Rightarrow$ keeps every entry of x away from 0.

Barrier primal QP

Step 1: replace inequality constraints by barrier

Replace the primal QP

$$\begin{array}{ll}\text{minimize} & c^T x + \frac{1}{2} x^T Q x \\ \text{subject to} & Ax = b \\ & x \geq 0\end{array}$$

with the barrier primal QP

$$\begin{array}{ll}\text{minimize} & c^T x + \frac{1}{2} x^T Q x - \mu \sum_{j=1}^n \ln x_j \\ \text{subject to} & Ax = b\end{array}$$

Logarithmic barrier and stationarity

Step 2: remove equality constraints using Lagrangian

$$\mathcal{L}(x, y, \mu) = c^\top x + \frac{1}{2} x^\top Q x - y^\top (Ax - b) - \mu \sum_{j=1}^n \ln x_j$$

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A stationary point (x, y, μ) of the Lagrangian satisfies

$$\nabla_x \mathcal{L}(x, y, \mu) = 0 \qquad = c + Qx - A^\top y - \mu X^{-1} e$$

with $X^{-1} = \mathbf{diag}(x_1^{-1}, \dots, x_n^{-1}) \in \mathbb{R}^{n \times n}, (x_j > 0)$.

KKT conditions for barrier problem

- Define $s := \mu X^{-1}e$, which implies $XS\mathbf{1} = \mu\mathbf{1}$, to get

KKT _{μ}

$$Ax = b$$

$$A^\top y + s - Qx = c$$

$$XS\mathbf{1} = \mu\mathbf{1}$$

$$(x, s) > 0$$

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KKT _{μ} $\rightarrow$ KKT as $\mu \rightarrow 0$.

Central path (LP case)

- ▶ Parameter μ controls the distance to optimality

$$c^\top x - b^\top y = c^\top x - x^\top A^\top y = x^\top s = n\mu$$

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- ▶ Analytic center (μ -center): unique point

$$(x(\mu), y(\mu), s(\mu)), \quad x(\mu) > 0, \quad s(\mu) > 0$$

that satisfies the KKT_μ conditions.

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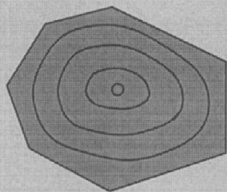
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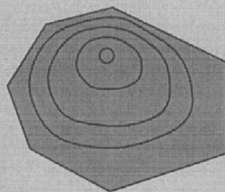
- ▶ The curve

$$\mathcal{C}_\mu = \{(x(\mu), y(\mu), s(\mu)) \mid \mu > 0\}$$

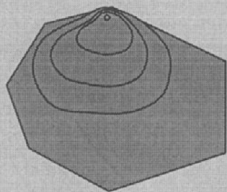
is called the primal-dual central path.



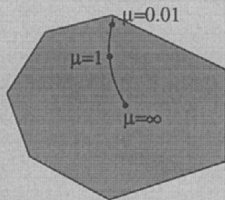
(a) $\mu=\infty$



(b) $\mu=1$



(c) $\mu=0.01$



(d) central path

Recall Newton's method for nonlinear equation

- For $F : \mathbb{R}^n \rightarrow \mathbb{R}^n$ smooth, solve $F(x) = 0$.

Recall Newton's method for nonlinear equation

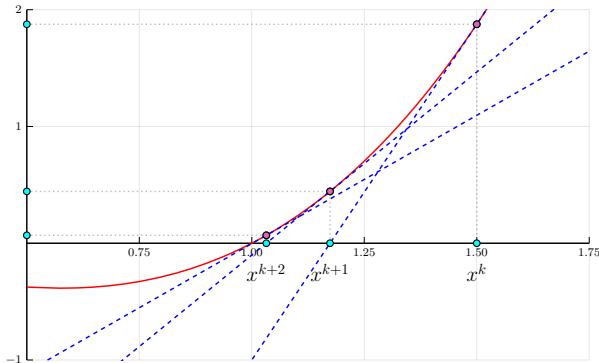
- ▶ For $F : \mathbb{R}^n \rightarrow \mathbb{R}^n$ smooth, solve $F(x) = 0$.
- ▶ Newton's method: define Jacobian $J_F(x)$ so $J_F(x)_{ij} = \frac{\partial F_i}{\partial x_j}$, and iterate

$$x^{k+1} = x^k - \alpha_k J_F(x^k)^{-1} F(x^k)$$

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Apply Method to KKT_μ

The first order optimality conditions for the barrier problem form a large system of nonlinear equations:

$$F(x, y, s) = 0,$$

where $F : \mathbb{R}^{2n+m} \mapsto \mathbb{R}^{2n+m}$ is defined as

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- ▶ The first two blocks are **linear**.
- ▶ The last block, corresponding to the complementarity condition, is **nonlinear**.
- ▶ Jacobian is

$$J_F(x, y, s) = \begin{bmatrix} A & 0 & 0 \\ -Q & A^\top & I \\ S & 0 & X \end{bmatrix}$$

Interior-point QP Algorithm

IPM Framework

Fix the barrier parameter μ and make *one* (damped) Newton step towards the solution of KKT_μ . Then reduce the barrier parameter μ and repeat.

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- ▶ Find step length α_k so $(x^k + \alpha_k \Delta x^k, y^k + \alpha_k \Delta y^k, s^k + \alpha_k \Delta s^k)$ is feasible.
- ▶ Make step $(x^{k+1}, y^{k+1}, s^{k+1}) = (x^k + \alpha_k \Delta x^k, y^k + \alpha_k \Delta y^k, s^k + \alpha_k \Delta s^k)$.

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Given $\epsilon > 0$, suppose that a feasible starting point $(x^0, y^0, s^0) \in \mathcal{N}_2(0.1)$ satisfies

$$(x^0)^\top s^0 = n\mu^0, \text{ where } \mu^0 \leq 1/\epsilon^\kappa,$$

for some positive constant κ . Then for some $K = \mathcal{O}(\sqrt{n} \ln(1/\epsilon))$, the optimality gap is bounded by ϵ after at most K iterations:

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- ▶ θ -neighborhood of the central path:

$$\mathcal{N}_2(\theta) := \{(x, y, s) \in \mathcal{F}^0 \mid \|XS\mathbf{1} - \mu\mathbf{1}\| \leq \theta\mu\}, \text{ with } \mu = \frac{1}{n}x^\top s.$$

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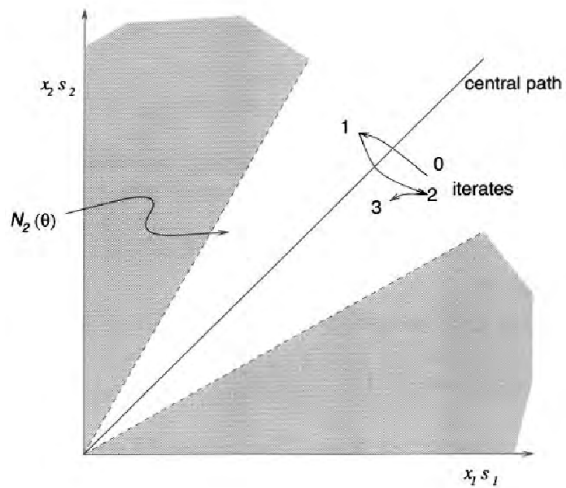
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- ▶ Slow progress towards optimality



Infeasible-start vs. feasible IPM

Feasible (path-following) IPM: keep iterates feasible.

Maintain $Ax = b$, $A^\top y + s - Qx = c$, $(x, s) > 0$ at every step and solve the Newton system

$$\begin{bmatrix} A & 0 & 0 \\ -Q & A^\top & I \\ S & 0 & X \end{bmatrix} \begin{bmatrix} \Delta x \\ \Delta y \\ \Delta s \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \sigma\mu\mathbf{1} - XS\mathbf{1} \end{bmatrix}, \quad \mu = \frac{x^\top s}{n}.$$

Only complementarity is perturbed; feasibility is preserved.

Infeasible-start IPM: allow and drive down feasibility residuals.

Start from any $(x > 0, s > 0, y)$ (not necessarily feasible) and solve

$$\begin{bmatrix} A & 0 & 0 \\ -Q & A^\top & I \\ S & 0 & X \end{bmatrix} \begin{bmatrix} \Delta x \\ \Delta y \\ \Delta s \end{bmatrix} = \begin{bmatrix} r_p \\ r_d \\ \sigma\mu\mathbf{1} - XS\mathbf{1} \end{bmatrix}, \quad \begin{aligned} r_p &= b - Ax, \\ r_d &= c + Qx - A^\top y - s. \end{aligned}$$

Infeasible start details

- ▶ *Direction decomposition.* Using linearity, separate computation of step into step to restore feasibility + step to improve complementarity: decompose $\Delta = \Delta_p + \Delta_d + \Delta_\mu$, where Δ_p, Δ_d restore feasibility and Δ_μ optimizes. In feasible IPM, $\Delta_p = \Delta_d = 0$.
- ▶ *Residual contraction.* Feasibility typically arrives before optimality, as linear system is easier to solve than nonlinear: with step sizes (α_P, α_D) ,

$$r_p^+ = (1 - \alpha_P) r_p, \quad r_d^+ = (1 - \alpha_D) r_d,$$

- ▶ *Positivity via fraction-to-the-boundary.* Choose

$$\alpha_P = \alpha_0 \max\{\alpha : x + \alpha \Delta x \geq 0\}, \quad \alpha_D = \alpha_0 \max\{\alpha : s + \alpha \Delta s \geq 0\}, \quad \alpha_0 \lesssim 1,$$

then update $x^+ = x + \alpha_P \Delta x$, $y^+ = y + \alpha_D \Delta y$, $s^+ = s + \alpha_D \Delta s$.

Augmented system

Newton direction

$$\begin{bmatrix} A & 0 & 0 \\ -Q & A^\top & I \\ S & 0 & X \end{bmatrix} \begin{bmatrix} \Delta x \\ \Delta y \\ \Delta s \end{bmatrix} = \begin{bmatrix} b - Ax \\ c - A^\top y - s + Qx \\ \mu^k \mathbf{1} - XS \mathbf{1} \end{bmatrix} =: \begin{bmatrix} \xi_p \\ \xi_d \\ \xi_\mu \end{bmatrix}$$

use last (complementarity) block to solve for Δs as a function of Δx .

Augmented system

Define $\Theta = XS^{-1}$ (ill-conditioned!). Then Δx and Δy solve the Newton system
 $\iff$

$$\begin{bmatrix} -Q - \Theta^{-1} & A^\top \\ A & 0 \end{bmatrix} \begin{bmatrix} \Delta x \\ \Delta y \end{bmatrix} = \begin{bmatrix} \xi_d - X^{-1} \xi_\mu \\ \xi_p \end{bmatrix}$$

- ▶ Newton system is nonsymmetric.
- ▶ Augmented system is symmetric but indefinite.

Normal equations

Augmented system

$$\begin{bmatrix} -\Theta^{-1} & A^\top \\ A & 0 \end{bmatrix} \begin{bmatrix} \Delta x \\ \Delta y \end{bmatrix} = \begin{bmatrix} \xi_d - X^{-1}\xi_\mu \\ \xi_p \end{bmatrix} \equiv: \begin{bmatrix} g \\ \xi_p \end{bmatrix}$$

Normal equations

Eliminate Δx to arrive at the *Normal equations*

$$(A\Theta A^\top)\Delta y = A\Theta g + \xi_p$$

Normal equations

Augmented system

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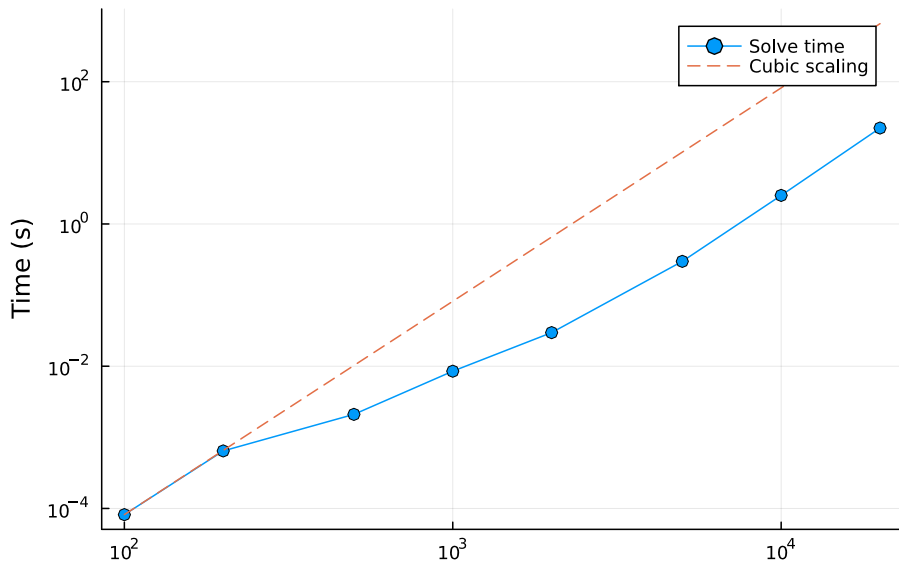
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$$(A\Theta A^\top)\Delta y = A\Theta g + \xi_p$$

- ▶ $A\Theta A^\top$ is symmetric and positive semidefinite. (Finally!)
- ▶ Normal equations in QP ($A(Q + \Theta)A^\top$) $\Delta y = g$ are generally nearly dense, even when A and Q are sparse.
- ▶ LP: Normal equations are often used.
- ▶ QP: usually use the indefinite augmented system.

Solving a linear system



IPMs in practice

demo:

Outline

IPM for linear and quadratic programs

IPM for convex nonlinear programming

IPM for conic optimization

IPM for NLP

► Convex NLP

$$\begin{array}{ll} \text{minimize} & f(x) \\ \text{subject to} & g(x) \leq 0 \end{array} \iff \begin{array}{ll} \text{minimize} & f(x) \\ \text{subject to} & g(x) + z = 0, \quad z \geq 0 \end{array}$$

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- Write out Lagrangian

$$L(x, y, z, \mu) = f(x) + y^\top (g(x) + z) - \mu \sum_{i=1}^m \ln(z_i)$$

IPM for NLP

- Write conditions for stationary point

$$\nabla_x L(x, z, y) = \nabla f(x) + J_g(x)^\top y = 0$$

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- Write KKT system

$$\nabla f(x) + J_g(x)^\top y = 0,$$

$$g(x) + z = 0$$

$$YZ\mathbf{1} = \mu\mathbf{1}$$

Newton for KKT of NLP

- ▶ Apply Newton method for KKT system

Newton for KKT of NLP

- ▶ Apply Newton method for KKT system
- ▶ Jacobian matrix of KKT system

$$J_F(x, z, y) = \begin{bmatrix} Q(x, y) & J_g(x)^\top & 0 \\ J_g(x) & 0 & I \\ 0 & Z & Y \end{bmatrix}$$

where $Q(x, y) = \nabla^2 f(x) + \sum_{i=1}^m y_i \nabla^2 g_i(x)$ is the Hessian of L

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- ▶ Newton step for KKT system

$$\begin{bmatrix} Q(x, y) & J_g(x)^\top & 0 \\ J_g(x) & 0 & I \\ 0 & Z & Y \end{bmatrix} \begin{bmatrix} \Delta x \\ \Delta y \\ \Delta z \end{bmatrix} = \begin{bmatrix} -\nabla f(x) - J_g(x)^\top y \\ -g(x) - z \\ \mu \mathbf{1} - YZ \mathbf{1} \end{bmatrix}$$

From QP to NLP

- Newton direction for NLP

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- ▶ Need to compute $Q(x, y)$ and $J_g(x)$ at each iteration
- ▶ Caveat: use trust region method to choose stepsize as Hessian may be indefinite.

Outline

IPM for linear and quadratic programs

IPM for convex nonlinear programming

IPM for conic optimization

Self-concordant function

Definition

Function $f : \mathbb{R} \rightarrow \mathbb{R}$ is *self-concordant* if for some constant $M_f \geq 0$, the inequality

$$f'''(x) \leq M_f |f''(x)|^{3/2}$$

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A function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is self-concordant if its restriction to any line is self-concordant. Equivalently,

$$\nabla^3 f(x)[u, u, u] \leq M_f \|u\|_{\nabla^2 f(x)}^{3/2}, \quad u \in \mathbb{R}^n$$

- ▶ A self-concordant function is always well approximated by a quadratic model.

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- ▶ A self-concordant function is always well approximated by a quadratic model.
- ▶ Self-concordance is invariant under affine transformations: if $g(z)$ is self-concordant, so is $f(x) = g(Ax - b)$

Newton's method converges quadratically for self-concordant functions

Recall we proved that Newton's method converges quadratically (locally) when the problem has Lipschitz Hessian (locally).

Using linesearch, a similar argument gives a *global* bound for self-concordant optimization:

Theorem ([Boyd and Vandenberghe, 2004, Section 11.5])

Newton's method with line search finds an ε approximate solution in less than $\text{constant} \times (f(x_0) - f^) + \log_2 \log_2 \frac{1}{\varepsilon}$ iterations.*

The constant depends only on the linesearch parameters c and β .

Barrier function candidates

Which of these functions is self-concordant? Strongly convex? Smooth?

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$f(x) = -\ln(x)$ is self-concordant in $\mathbb{R}_+$ because

$$f'(x) = -\frac{1}{x}, \quad f''(x) = \frac{1}{x^2}, \quad f'''(x) = -\frac{2}{x^3}$$

$f(x) = \exp(1/x)$ is not.

Conic optimization

- Consider the optimization problem

$$\begin{array}{ll}\text{minimize} & c^\top x \\ \text{subject to} & Ax = b \\ & x \in K\end{array}$$

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$$c^\top x - b^\top y = x^\top (c - A^\top y) = x^\top s \geq 0$$

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- Conic optimization can be solved in polynomial time with IPMs

Second-order conic optimization

- ▶ $\mathcal{K}_{\text{SOC}} := \{(x, t) \mid x \in \mathbb{R}^{n-1}, t \in \mathbb{R}, \|x\|_2 \leq t, t \geq 0\}$ (Second-order cone)

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- ▶ Logarithmic barrier function for the second-order cone

$$f(x, t) = \begin{cases} -\ln(t^2 - \|x\|_2^2) & \text{if } \|x\| < t \\ +\infty & \text{otherwise} \end{cases}$$

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Theorem

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Exercise: Prove in case $n = 2$.

Semidefinite programming

- ▶ Variable now is a symmetric matrix $X \in K = \mathbf{S}^n$

SDP and its dual

$$\begin{array}{ll}\text{minimize} & C \bullet X \\ \text{subject to} & A_i \bullet X = b_i, i = 1, \dots, m \\ & X \succeq 0\end{array}$$

$$\begin{array}{ll}\text{maximize} & b^\top y \\ \text{subject to} & \sum_{i=1}^m y_i A_i + S = C \\ & S \succeq 0\end{array}$$

$A_i, C \in \mathbf{S}^n$ and $b \in \mathbb{R}^m$ given, and $X, S \in \mathbf{S}^n$ and $y \in \mathbb{R}^m$ unknown.

Semidefinite programming

- ▶ Variable now is a symmetric matrix $X \in K = \mathbf{S}^n$
- ▶ Define $X \bullet Y = \text{tr}(X^\top Y)$

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Theorem (Weak duality for SDP)

If X is primal feasible and (y, S) is dual feasible, then

$$C \bullet X - b^\top y = X \bullet S \geq 0$$

Logarithmic barrier for SDP

- Logarithmic barrier function for the semi-definite cone

$$f(X) = \begin{cases} -\ln(\det(X)) & \text{if } X \succ 0 \\ +\infty & \text{otherwise} \end{cases}$$

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 - ▶ $\ln(1 + t \operatorname{tr}(U)) \approx t \operatorname{tr}(U)$
- ▶ Let $X \succ 0$ and $H \in \mathbf{S}^n$ be given. Then

$$\begin{aligned} f(X + tH) &= -\ln(\det(X + tH)) = -\ln(\det(X(I + tX^{-1}H))) \\ &= -\ln(\det(X)) - \ln(\det(I + tX^{-1}H)) \\ &= -\ln(\det(X)) - \ln(1 + t \operatorname{tr}(X^{-1}H) + \mathcal{O}(t^2)) \\ &= f(X) - tX^{-1} \bullet H + \mathcal{O}(t^2) \end{aligned}$$

Derivatives of Logarithmic barrier for SDP

- First derivative of $f(X)$

$$f'(X) = \lim_{t \rightarrow 0} \frac{f(X + tH) - f(X)}{t} = -X^{-1}$$

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- Second derivative of $f(X)$

$$\begin{aligned} f'(X + tH) &= -[X(I + tX^{-1}H)]^{-1} = -[I - tX^{-1}H + \mathcal{O}(t^2)]X^{-1} \\ &= f'(X) + tX^{-1}HX^{-1} + \mathcal{O}(t^2) \end{aligned}$$

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so $f''(X)[H] = X^{-1}HX^{-1}$ and $D^2f(X)[H, G] = X^{-1}HX^{-1} \bullet G$.

- $f'''(X)[H, G] = -X^{-1}HX^{-1}GX^{-1} - X^{-1}GX^{-1}HX^{-1}$

Characterization of self-concordance for SDP

Theorem

The function $f(X) = -\ln \det X$ is a convex barrier for $\mathbf{S}_+^n$.

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Proof sketch.

Let $\varphi(t) = F(X + tH)$. Then, prove that $\varphi''(t) \geq 0$ for $t > 0$ such that $X + tH \succ 0$. Therefore, when $X \succ 0$ approaches a singular matrix, its determinant approaches zero, and the function $f(X) \rightarrow +\infty$. □

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Theorem ([Nestervov and Nemirovskii, 1994])

The barrier function $f(X) = -\ln \det X$ is self-concordant on $\mathbf{S}_+^n$.

Solving SDPs with IPMs

- Replace the primal SDP

$$\begin{aligned} &\text{minimize} && C \bullet X \\ &\text{subject to} && \mathcal{A}X = b, \\ &&& X \succeq 0, \end{aligned}$$

with the primal barrier SDP

$$\begin{aligned} &\text{minimize} && C \bullet X + \mu f(X) \\ &\text{subject to} && \mathcal{A}X = b, \end{aligned}$$

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- Formulate the Lagrangian

$$L(X, y, S) = C \bullet X + \mu f(X) - y^T (\mathcal{A}X - b),$$

with $y \in \mathcal{R}^m$, and write the first order conditions (FOC) for a stationary point of L :

$$C + \mu f'(X) - \mathcal{A}^* y = 0$$

Solving SDPs with IPMs (cont'd)

- ▶ Use $f(X) = -\ln \det X$ and $f'(X) = -X^{-1}$ to obtain

$$C - \mu X^{-1} - \mathcal{A}^* y = 0$$

Solving SDPs with IPMs (cont'd)

- ▶ Use $f(X) = -\ln \det X$ and $f'(X) = -X^{-1}$ to obtain

$$C - \mu X^{-1} - \mathcal{A}^* y = 0$$

- ▶ Denote $S = \mu X^{-1}$, i.e., $XS = \mu I$. Then, the FOC can be written as

$$\mathcal{A}X = b$$

$$\mathcal{A}^* y + S = C$$

$$XS = \mu I$$

with $X, S \in \mathbf{S}_{++}^n$.

Newton direction

Differentiating this system is hard! The Newton direction solves:

$$\begin{bmatrix} \mathcal{A} & 0 & 0 \\ 0 & \mathcal{A}^* & \mathcal{I} \\ \mu(X^{-1} \odot X^{-1}) & 0 & \mathcal{I} \end{bmatrix} \cdot \begin{bmatrix} \Delta X \\ \Delta y \\ \Delta S \end{bmatrix} = \begin{bmatrix} \xi_b \\ \xi_C \\ \xi_\mu \end{bmatrix}.$$

We define the Kronecker product $P \odot Q$ for $P, Q \in \mathbb{R}^{n \times n}$, which yields a linear operator from $\mathbf{S}^n$ to $\mathbf{S}^n$ given by

$$(P \odot Q)U = \frac{1}{2} (PUQ^T + QUP^T).$$

Summary

- ▶ IPM for SOCP and SDP with self-concordant barrier:

source: [Gondzio, 2012: Interior Point Methods 25 Years Later]

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- ▶ IPM for SOCP and SDP with self-concordant barrier:
 - ▶ polynomial complexity (predictable behaviour)
- ▶ Unified algorithm with fast convergence
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- ▶ efficient for LP, QP, SOCP

source: [Gondzio, 2012: Interior Point Methods 25 Years Later]

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 - ▶ and this requires n^6 flops!

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